Databricks (Attunity Ingestion)

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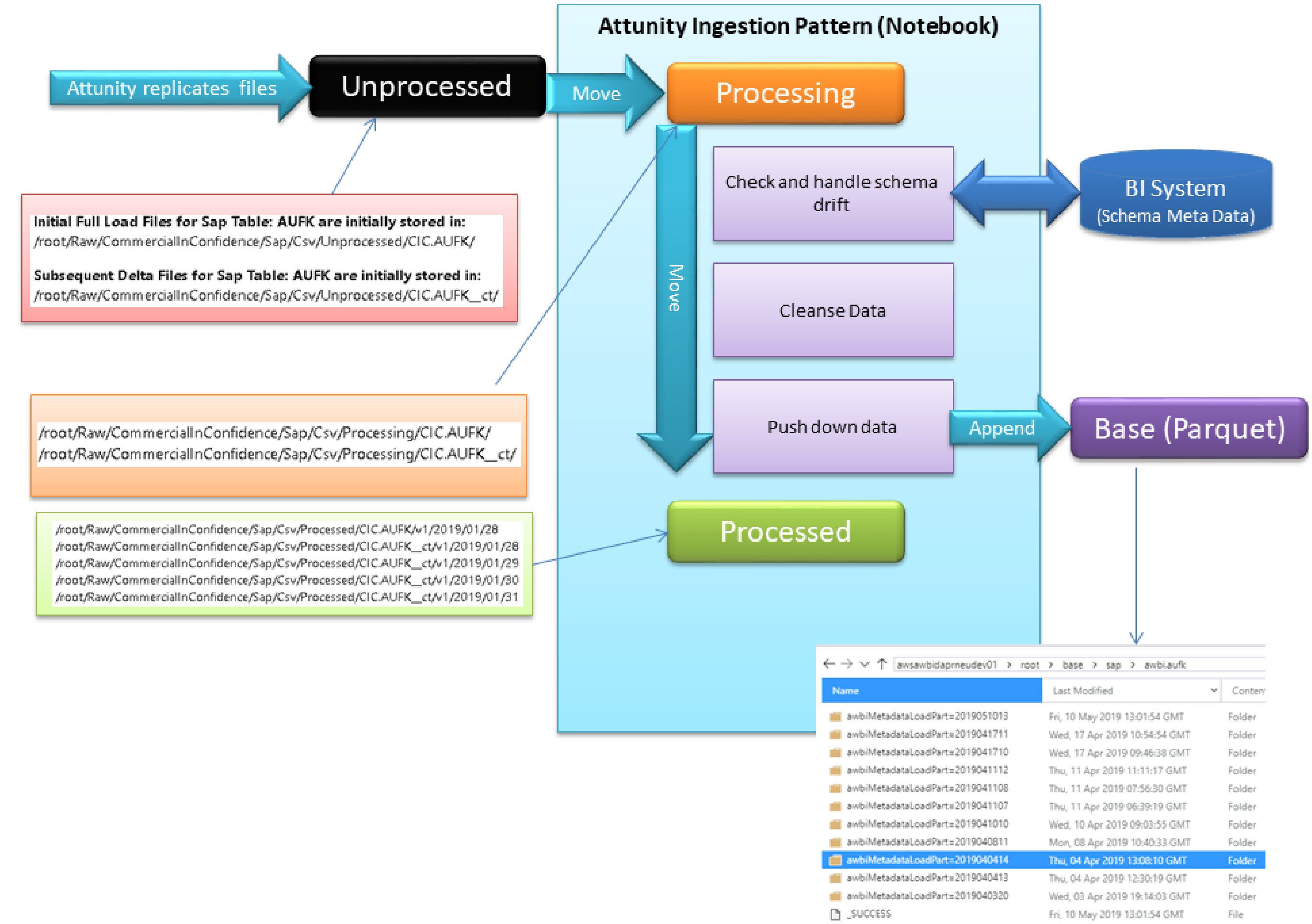
Databricks Attunity Ingestion Notebook

This document aims to describe the functionality of the Databricks Attunity ingestion notebook and provide guidance for development community on its usage. The following illustration provided an overview of the end to end process:



Purpose

The sole purpose of this notebook is to provide processing functionality for the data that lands in "Raw" within the Attunity DataLake. The notebook described here lifts the data landed in "Raw" and processes in a consistent manner into "Base".

Description

The Databricks Attunity Ingestion notebook is designed to be a single notebook that will process all entities listed in BISystem as data source provided by SAP Attunity. Azure Data Factory will acquire a list of entities from BISystem and in-turn, call the notebook for processing into "Base".

Parameters

The only parameter is the entity ID (entity_id). When automated via Azure Data Factory, this will be picked up as described above from BISystem, however, when testing, a developer will be required to manually run the Azure Data Factory Pipeline <ENTER PIPELINE NAME HERE> and manually feed in the required entity ID.

Development Process

When running the notebook for the first time, the notebook will attempt to gather all the require entity's column meta data and automatically insert into BISystem, creating Version 1 of the entity column meta data. As schema drift occurs over time, the parquet files in the datalake will handle this and the structure will continue to be versioned accordingly within the AWBISystem database.

Processing Data in Raw

In order to ensure all files are successfully processed into "Base", some reorganisation of the files contained within "Raw" must occur first. As an example, data will be landed from Attunity into the following folder within the Azure Data Lake Store (ADLS);

- adl://awsawbiattunityneudev01.azuredatalakestore.net/root/raw/sap/json/**Unprocessed**/SAP.QMEL/Load0000001.json

Before processing into "Base", this and all other files in this directory are moved into the following location;

- adl://awsawbiattunityneudev01.azuredatalakestore.net/root/raw/sap/json/**Processing**/SAP.QMEL/Load0000001.json

Once the data has been processed into "Base", the file is then moved from the above location into the following;

- adl://awsawbiattunityneudev01.azuredatalakestore.net/root/raw/sap/json/**Processed**/SAP.QMEL/Load0000001.json

Initial and Change Tracking (ct)

With Attunity an initial snapshot is taken of how the data looks at the point in time when first activated within the Attunity program, this always lands into a ADLS folder that corresponds to the above example. However, subsequent transactional changes are tracking into a separate folder, illustrated below;

- adl://awsawbiattunityneudev01.azuredatalakestore.net/root/raw/sap/json/Unprocessed/**SAP.QMEL__ct**/Load0000001.json

These files are processed in the same way, moving across an "Unprocessed" and "Processed" collection of files. The only difference between these two sets of data is that Attunity will inject header meta data into the change tracking files.

The structure of the data is extracted from the DFM (Delphi Form) files, each json file that is exported by Attunity is coupled with an associated DFM file. If for example, a change of schema occurs at source, then it could be possible that there are multiple versions of schema sat in the "unprocessed" folder. The Parquet format which is currently used as the target in the data lake for the base and curated layers handle this schema drift and align columns accordingly by name.

Changing Entity Column Versions

If the above scenario occurs described towards the end of the "Initial and Change Tracking" section then a new version of the entity columns will need to be created. This is automatically handled.

Data Cleansing

The main value add of the "Base" layer within ADLS is data cleansing, data should be consistently cleansed and ready for data processing. The data cleansing actions that occur within the notebook are described below;

- Trimming spaces from left and the right of text
- Replacing a single space column (common SAP occurrence) with a '?'
- Replacing null with '?'
- Dropping duplicate rows

Bad Data

Bad data is not processed within the Attunity notebook.